Receiver-Conditioned Latent Uncertainty Propagation for Multi-Agent Web Reasoning

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Abstract

Role-specialized language-model agents retrieve, interpret, and combine Web evidence through dependent intermediate judgments. An erroneous premise can therefore propagate with misleading confidence, while uncertainty over a sender's artifact need not describe the receiver's task. We present Receiver-Conditioned Latent Uncertainty Propagation (RCLUP), which associates alternative artifacts with latent content and provenance, reads them jointly in the receiver's task, and selects source-grounded inter-agent clarifications. The carrier is trained on both commitment and clarification outcomes, with an action-profile objective that preserves distinctions among questions rather than only predicting continuation failure. Fixed reference executors make representation targets stationary; after freezing the receiver, clarification risks are retargeted to its actual post-reply decisions. Role-conditional calibration supplies explicit final-error semantics for action scores, while a separate reporting estimator describes the frozen deployed policy and its actual final answer. Returned evidence revises versioned messages and dependent decisions. The evaluation separates inherited-error propagation, cross-role calibration, and clarification utility against rich-text, generic-latent, and intervention-trained controls. Numerical results are left unreported in this pre-experimental manuscript.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Question answering**.

Keywords

multi-agent systems, uncertainty propagation, calibration, latent communication, clarification, Web question answering

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1 Introduction

Web question answering requires more than locating a relevant page. A system must identify the intended entity, interpret qualifications in source material, combine evidence across documents, and verify that the resulting answer follows from that evidence. Role-specialized large language model (LLM) agents divide these operations among retrievers, interpreters, reasoners, and verifiers. Their outputs are not independent votes on one answer: a selected document becomes an interpreter's input, and an extracted claim becomes a reasoner's premise. Consequently, an error can travel beyond the role that introduced it, while repeated use of the same source can make dependent judgments appear corroborative. Reliable multi-agent Web reasoning therefore requires uncertainty to remain meaningful as evidence crosses these interfaces.

Two problems arise at these handoffs. First, an incorrect premise can propagate with unwarranted confidence, particularly when downstream agents reuse evidence inherited from a common ancestor. Propagation-aware risk estimation and communication-dependent calibration address parts of this problem [5, 7]. Second, the sender and receiver often predict different events. Uncertainty over an entity identity is not uncertainty over a temporal comparison derived from that entity's attributes. Even a correctly calibrated sender score does not determine its relevance to the receiver. A disagreement between 1998 and 2001 matters when deciding whether an event preceded 2000, but not when both alternatives imply the same answer to the receiving task. Propagation must therefore preserve the alternatives behind a judgment and interpret them in the receiver's actual task, rather than merely rescale local confidence.

A calibrated downstream risk still does not identify the appropriate clarification. Two handoffs can be equally likely to fail while requiring different questions: one calls for distinguishing homonymous entities, another for distinguishing an incorporation date from the start of operations. Failure prediction and intervention value are distinct objectives [11]. A representation trained only to predict failure can consequently be useful for rejection while discarding distinctions needed to choose a question. Effective propagation requires an uncertainty-bearing message that supports both a well-defined risk forecast and the selection of a specific, source-grounded clarification.

Continuous communication provides a means of retaining information beyond a committed answer. CIPHER transmits probability-weighted embeddings, LatentMAS shares latent thoughts, and Interlat learns receiver-side adaptation and task-oriented compression [2, 8, 13]. Interlat also preserves uncertainty-reducing behavior, so neither receiver awareness nor uncertainty-aware compression is new in itself. The question here is narrower: how should candidate-associated content be transmitted and trained when different receivers need to resolve different distinctions? A generic latent payload, a scalar risk, and a structured textual explanation can all convey useful information; their value for selecting a clarification must be distinguished under the same supervision and interaction budget.

We introduce *Receiver-Conditioned Latent Uncertainty Propagation* (RCLUP) for this setting. Each sender transmits candidate artifacts, empirical support, compressed candidate-specific hidden states, and explicit provenance. The receiver jointly reads these messages with its own task and their evidence dependencies, then estimates the final-error risk of each admissible commitment. The same representation scores concrete questions directed to a sender and a candidate distinction. Crucially, the carrier is trained using both commitment outcomes and clarification outcomes, with an

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additional objective preserving differences between action values. Thus, information useful only for choosing a clarification receives a direct training signal rather than being expected to survive a failure-only bottleneck.

To keep this learning well-defined, representation training uses fixed reference continuations. After the carrier and commitment rule are frozen, clarification risks are re-estimated under the actual post-reply receiver; role-conditional calibration then supplies comparable probability semantics for control. Source-backed replies revise the versioned messages and trigger recomputation of affected decisions. A separate, reporting-only estimator is calibrated on trajectories of the frozen closed-loop policy, so confidence attached to a final answer describes that answer rather than an earlier, hypothetical continuation. The method uses latent content to select artifacts and clarifications; it does not claim unrestricted latent question generation or a new definition of information value.

Our contribution couples candidate-indexed communication with task- and dependency-aware reception, trained directly for clarification decisions. The evaluation isolates three claims: whether inherited errors persist less often, whether final-error forecasts are reliable across heterogeneous roles, and whether candidate-specific latents improve useful clarification beyond rich text and generic latent communication. We study these questions in multi-source Web evidence navigation, where retrieving, interpreting, and verifying the same evidence are dependent operations, and include discrete reasoning as a transfer setting. Numerical outcomes are intentionally left unreported in this pre-experimental manuscript.

2 Related Work

Uncertainty across agent dependencies. PropUQ-MAS combines local risk with inherited risk and a receiver's acceptance tendency, using conditional-independence assumptions for tractable propagation [7]. CAGE-Cal uses communicating and counterfactual non-communicating graphs to calibrate panel confidence [5]. These methods establish that communication affects reliability. Our setting additionally requires different receiver tasks to identify which upstream distinction merits clarification. We retain provenance and jointly read candidate content rather than infer independence from agreement. Dependency features are not themselves a guarantee of correct reasoning.

Latent communication. Hidden-activation probes predict statement truthfulness, while semantic entropy probes target semantic uncertainty [1, 6]. CIPHER, LatentMAS, and Interlat use continuous representations for inter-agent communication [2, 8, 13]. In particular, Interlat combines task utility, uncertainty-weighted behavior preservation, and latent alignment. Our distinction is not merely adding a learned receiver: candidate associations and a clarification-specific action profile are the representation choices under test. Matched generic-latent and rich-text receivers receive the same outcome supervision.

Calibration and information-seeking control. Post-hoc calibration adjusts predictive probability semantics [4], and AUQ retains uncertainty explanations to guide reflection [12]. Intervention-value learning separates failure detection from beneficial control [11]. We adopt this distinction, not its novelty. RCLUP trains communication to preserve differences among concrete, evidence-addressable questions, retargets its scores when the reply executor changes, and separately evaluates reference-action calibration and deployed-policy calibration.

3 Problem Formulation

An execution is a time-unrolled DAG $G = (V, E)$ whose nodes are invocations, not persistent agent identities. Invocation i has role r_i, actual subtask T_i, accessible evidence D_i, and public history H_i. It produces an artifact in a role-specific space: ranked documents, fact tuples, derived answers, or verification verdicts. A later clarification creates a new invocation and message version. The receiver state S_j contains its task, evidence, incoming packets, graph metadata, and remaining budgets. Gold labels are available only to the offline evaluator.

We separate three quantities. *Candidate support* summarizes the sender's sampled proposals, not their correctness. *Action risk* predicts the final answer's error under a specified continuation. *Clarification value* compares the outcomes of asking and committing, after charging the question. None is obtained by normalizing entropies from different local output spaces.

Let π_0 be a fixed ordinary workflow with the shared backbone, tools, verification, and repair. It contains no learned handoff controller, but retains normal evidence checking. For a receiver artifact b, define

$$r_j^0(b) = \Pr(E_{j,b}^{\pi_0} = 1 \mid S_j, \mathrm{commit}(b)), \tag{1}$$

where $E = 1$ means the eventual answer fails the native correctness validator; missing or unresolved final answers also fail. Several intermediate artifacts may all yield success, so their risks are not an answer posterior. Local correctness is a separate diagnostic when native labels support it.

For a question q, the post-reply executor must also be specified. We use $r_j^0(q)$ when the reply is followed entirely by π_0, and $r_j^*(q)$ when the reply is followed by the frozen learned commitment rule and then π_0. These are reference-action forecasts. Deployed-policy risk r_j^{pol} instead predicts final error under the complete frozen controller, including later clarifications. These targets are never interchanged when scoring revised outputs.

4 Method

4.1 Candidate-Indexed Uncertainty Carriers

A frozen backbone draws S stochastic proposals and one greedy proposal from (r_i, T_i, H_i, D_i). A shared, role-specific canonicalizer groups equivalent artifacts. We retain the greedy artifact and the most frequent alternatives, up to $K_i \le K$ distinct candidates; all other stochastic draws, including invalid ones, enter a residual category $\perp$. With counts n_{ik} and pseudocount α,

$$w_{ik} = \frac{n_{ik} + \alpha}{S + \alpha(K_i + 1)}, \qquad w_{i\perp} = \frac{S - \sum_k n_{ik} + \alpha}{S + \alpha(K_i + 1)}. \tag{2}$$

Only stochastic draws contribute counts. The residual represents discarded sampled mass, not the probability of all unseen correct answers.

For candidate k, the representative generation with greatest length-normalized log probability supplies response hidden states

$H_{ik} \in \mathbb{R}^{L_{ik} \times d_B}$. A trainable attention pooler produces m slots:

$$Z_{ik}^{\text{loc}} = \text{softmax}\left(\frac{Q(H_{ik}W_K)^\top}{\sqrt{d}}\right) H_{ik}W_V \in \mathbb{R}^{m \times d}. \tag{3}$$

A packet $M_i = \{(a_{ik}, w_{ik}, Z_{ik}, \mathcal{P}_{ik})\}_k$ also identifies its selected artifact and version. Provenance $\mathcal{P}_{ik}$ records source spans and consumed origin/version identifiers. The residual has learned null slots and a missing-content flag, with no invented source. Root packets use $Z_{ik} = Z_{ik}^{\text{loc}}$; subsequent packets use Equation (6). Hidden states can retain evidence processed during generation but are not evidence certificates. All matched controls receive the same readable candidates, support weights, and provenance.

4.2 Task-Conditioned Reading with Dependence Information

The receiver encodes its role, actual subtask, evidence, and readable history as $s_j \in \mathbb{R}^d$. The frozen decoder proposes a set $\mathcal{B}_j$ of role-valid artifacts, including an unresolved option. Latents do not enter the question or artifact proposer: they guide the learned selection and subsequent communication. This fixes proposal coverage across matched comparisons.

Exactly identical origin/version/content packets are coalesced. Remaining candidate slots x_u include candidate-text, role, slot-position, and missingness encodings. Two joint attention blocks use

$$A_{uv} = \frac{(x_u W_Q)(x_v W_K)^\top}{\sqrt{d}} + \eta^\top d_{uv}, \tag{4}$$

where d_{uv} contains origin identity, directed ancestry, ancestor overlap, source-span overlap, and a missing-provenance mask. This produces contextual slots h_u. The reader learns from joint outcomes; it does not multiply incoming correctness probabilities.

We additionally normalize the contribution of each packet before combining packets. For packet i and receiver candidate b,

$$\begin{aligned}
t_{jb} &= W_t[s_j; \text{enc}(b)], \\
\beta_{jbiu} &= \text{softmax}_{u \in i}\left(t_{jb}^\top W_k h_u / \sqrt{d} + \log w_u - \log m\right), \\
c_{jbi} &= \sum_{u \in i} \beta_{jbiu} W_v h_u, \\
\rho_{jbi} &= \text{softmax}_i\left(f_\eta(s_j, \text{enc}(b), c_{jbi}, d_i) - \log \nu_i\right), \\
c_{jb} &= \sum_i \rho_{jbi} c_{jbi}.
\end{aligned} \tag{5}$$

Here w_u is its candidate's support, d_i summarizes that packet's relations, and $\nu_i = 1 + \sum_{p \neq i} o_{ip}$. The overlap o_{ip} is the maximum of source-span and evidence-bearing-ancestor Jaccard overlap; routing-only ancestors are excluded. Empty or missing sets contribute zero overlap and retain a missingness flag. The $1/\nu_i$ prior reduces the multiplicity advantage of correlated packets without discarding their content. It is a redundancy prior, not an independence certificate; genuine new evidence remains available through the contextual slots and learned gate.

With graph summaries v_j, define $\xi_{jb} = [s_j; \text{enc}(b); c_{jb}; v_j]$. The commitment logit $\ell_j(b) = f_\theta(\xi_{jb})$ predicts Equation (1). Dependence changes the features used to assess and choose artifacts, whereas calibration only changes the probabilistic interpretation of a fixed score.

Recursive propagation. When receiver j becomes a sender, it constructs the outgoing slots for each candidate b as

$$\begin{aligned}
\gamma_{jb} &= \text{sigmoid}(W_g[s_j; \text{enc}(b); c_{jb}] + b_g), \\
Z_{jb,t}^{\text{out}} &= \text{LN}\left(Z_{jb,t}^{\text{loc}} + \gamma_{jb} \odot W_o c_{jb}\right), \quad t = 1, \ldots, m.
\end{aligned} \tag{6}$$

All consumed origins remain in provenance, independent of attention weight. This carries inherited content across multiple tasks without treating an inherited representation as a fresh independent observation. Re-encoding can still introduce errors; dependence-aware training and the propagation evaluation test that limitation directly.

4.3 Clarification-Supervised Representation Learning

A carrier optimized only for commitment risk need not preserve information that distinguishes two equally risky but differently repairable handoffs. We therefore train the carrier on an *action profile*: the outcomes of its candidate commitments and concrete clarifications.

A common frozen proposer creates at most $Q_{\max}$ atomic questions from the public task, readable candidates, source identifiers, and previous questions. Each $q = (i, k, k', \text{text}, \text{scope})$ addresses a sender and candidate distinction. Candidate-versus-residual questions are allowed. A schema checker removes compound or unaddressable questions before any outcomes are seen. The menu is identical across methods at the same public state; menu coverage is evaluated separately.

The question features ζ_{jq} concatenate s_j, $\text{enc}(q)$, the targeted candidate means $\bar{h}_{ik}, \bar{h}_{ik'}$, their difference and absolute difference, support weights, receiver-attention summaries, and v_j. A reference head $g_{\kappa_0}(\zeta_{jq})$ predicts $r_j^0(q)$. In fitting data, each candidate is committed and each admitted question is actually answered from accessible sources, followed by the fixed π_0. Paired, predetermined seeds produce binary outcomes e_{jur}^0 and means $\bar{e}_{ju}^0$. The executor never consults the currently trained carrier, so changing its parameters cannot change these targets.

Let $p_j^0(u)$ be the sigmoid output of the appropriate commitment or reference-question head, and let $\mathcal{U}_j = \mathcal{B}_j \cup \mathcal{Q}_j$. For each original question, average losses over its prefixes and actions:

$$\begin{aligned}
\mathcal{L}_C &= \mathbb{E}_{j,b,r}\, \text{BCE}(p_j^0(b), e_{jbr}^0), \\
\mathcal{L}_Q &= \mathbb{E}_{j,q,r}\, \text{BCE}(p_j^0(q), e_{jqr}^0), \\
\mathcal{L}_\Delta &= \mathbb{E}_{j,u<v}\left[(p_j^0(u) - p_j^0(v)) - (\bar{e}_{ju}^0 - \bar{e}_{jv}^0)\right]^2, \\
\mathcal{L}_{\text{repr}} &= \mathcal{L}_C + \mathcal{L}_Q + \mu \mathcal{L}_\Delta.
\end{aligned} \tag{7}$$

Commit and question terms are normalized separately; empty question menus contribute zero question loss. The difference term includes question–question and question–commit pairs and retains noisy magnitudes rather than inventing strict preference labels. Cost is applied at decision time, not confused with error probability. *Both question terms backpropagate into the pooler, dependency reader, and recursive carrier*, as well as the heads. This is the critical departure from freezing a failure-only representation before learning clarification. Action-value supervision is established prior work [11]; the mechanism tested here is its use to preserve receiver-specific candidate distinctions in the communication bottleneck.

4.4 Executor-Matched Risks and Calibration

Joint training uses a fixed executor, while deployment chooses artifacts with the learned receiver. To avoid silently changing the meaning of question labels, freeze θ^* and define

$$b^*(S) = \arg\min_{b \in \mathcal{B}(S)} \ell_{\theta^*}(S, b). \tag{8}$$

For every fitting question, execute its real reply, rebuild the state S', commit $b^*(S')$, and then run π_0. These outcomes define $r_j^*(q)$. Initialize a deployment question head g_κ from g_{κ_0} and fit it to these labels, updating *only that head*. The carrier already contains clarification supervision from Equation (7); the second pass matches the action predictor to the actual reply executor without moving its targets. Reference heads are retained only for fixed-executor diagnostics.

On disjoint action-calibration questions, fit a positive-slope, partially pooled logistic map:

$$\widehat{r}_j(u) = \mathrm{sigmoid}\big(e^{a_0 + a_g} \ell_j(u) + b_0 + b_g\big), \quad g = (r_j, \mathrm{family}(u)). \tag{9}$$

The family distinguishes commit and ask; the corresponding logits are f_{θ^*} and g_κ. Bernoulli NLL plus a fixed L_2 penalty fits group deviations; sparse or unseen groups use the global map. A common positive slope within the commit group leaves Equation (8) unchanged, keeping the post-reply executor fixed through calibration. The scores now refer to a common final-error event, not a common local entropy scale. Selected-action reliability is audited separately because marginal calibration need not survive action selection.

4.5 Budgeted Clarification and Evidence Revision

The estimated net benefit of an admitted question is

$$\widehat{G}_j(q) = \widehat{r}_j(\mathrm{commit}(b^*)) - \widehat{r}_j(\mathrm{ask}(q)) - \lambda c(q). \tag{10}$$

We ask the maximum-positive-gain question that fits the remaining budgets, otherwise commit b^*. One atomic information item has $c(q) = 1$; inference cost is separately constrained. Ties favor fewer questions and then a fixed candidate order. This is a one-step reference-value controller, not an optimal multi-step information-gathering policy.

The sender answers from its authorized evidence and tools, returning a direct answer, source spans, version identifiers, and an unresolved flag. Quotation checks establish provenance, not truth. New evidence generates a new sender packet and invalidates descendants that consumed the previous version. Those descendants are recomputed in topological order, retaining independent evidence; a reply is not automatically accepted. Duplicate evidence-event identifiers are idempotent, and the same unresolved question is not repeated against an unchanged source version. The common verifier and normal repair remain active. Finite query, revision, and inference budgets ensure termination without discarding failures.

4.6 Calibration of the Executed Policy

Repeated application of Equation (10) defines a frozen deployed policy π^* that can clarify again at later handoffs. Its actual risk need not equal a forecast followed by π_0. We therefore fit a *reporting-only* head h_ψ on completed trajectories of π^* from a separate fitting partition. At each visited handoff it reads only the pre-action state, selected action, and frozen latent features; at termination it reads the actual final answer and final state. Every record is labeled by that trajectory's final error. A second disjoint calibration partition fits Equation (9), grouped by role and reporting phase.

These scores estimate $r_j^{\mathrm{pol}} = \Pr(E_{\mathrm{final}}^{\pi^*} = 1 \mid S_j, u_j)$ and the final-answer risk. They never feed back into control. Thus policy calibration cannot change answers, and neither a pre-clarification risk nor a reference-policy forecast is attached to a revised final answer. Each budget configuration is frozen and calibrated separately. All fits weight original questions equally rather than treating correlated handoffs as independent tasks.

Algorithm 1 Jointly trained propagation and calibrated clarification

Require: Frozen modules, source snapshot, role graph, budgets
for each invocation j in execution order **do**
  Build shared candidate/menu proposals from public history
  Canonicalize packets; jointly read task and dependencies
  Compose candidate latents and compute commit/ask risks
  if a positive-gain question is admissible **then**
    Obtain source-backed reply; charge all work
    Version evidence; invalidate and requeue affected nodes
  else
    Submit b^* to common verification and ordinary repair
    Emit checked artifact, or unresolved, and its carrier
  end if
  Log selected-action policy-risk forecast without feedback
end for
Report actual final answer with its post-update calibrated risk

5 Experiments

The protocol below is fixed for the proposed method. Numerical cells are left as — (not measured); no empirical improvement is asserted.

5.1 Tasks and Functionally Different Agents

HotpotQA and MuSiQue provide multi-hop questions and supporting evidence; DROP supplies a complementary discrete-reasoning setting [3, 9, 10]. The primary Web experiments operate on released document snapshots with reproducible retrieval. They test source selection, interpretation, cross-document composition, and evidence checking, rather than merely applying different personas to a closed-book answer. DROP uses its supplied passage and calculator as a reasoning-transfer test, not a Web retrieval claim.

A fixed planner uses only the public question to create at most three retrieval–interpretation branches, followed by reasoning and verification. This produces chains and branch–merge graphs with naturally shared evidence. All methods use the same BM25 index, direct source fetching, calculator, necessary history, and two ordinary verifier-repair rounds. Receivers can reread sources directly. No erroneous premise is injected into the main task comparison, and gold support or decomposition never guides agent retrieval. Table 1 specifies the role interfaces.

Table 1: Functional roles and their different artifact spaces.

Role	Operation	Outgoing artifact
Retriever	Retrieve evidence for a sub-question	Ranked document IDs and passages
Interpreter	Resolve entities, dates, or relations	Fact tuples with source spans
Reasoner	Compare or combine extracted facts	Derived answer and dependencies
Verifier	Check claims, sources, and calculations	Verdict and repair request

Qwen3-8B and Mistral-7B-Instruct-v0.3 are evaluated separately, with one frozen backbone shared across roles in each run. Thus heterogeneous tasks are not confounded with cross-model latent alignment. Official training questions are partitioned by root question and available composition family into representation fitting, policy-risk fitting, selection, action calibration, and policy calibration in a 55:15:10:10:10 ratio. Outcome-independent hashes select the fixed collection sizes in Appendix B. All branches and paraphrases stay with their root. The official labeled development split is held out for evaluation, not tuning. A separate source-disjoint evaluation removes overlapping training sources before fitting, without deleting unfavorable test outcomes.

5.2 Strong Controls and External Methods

Text MAS runs the complete ordinary workflow. The *scalar controller* receives calibrated local and downstream risk, task, and role features and is trained on action outcomes rather than only thresholded failure risk. *Rich text* includes every candidate, empirical weight, provenance, source span, and a structured explanation of its evidence conflict or ambiguity. *Generic latent* pools all candidate traces jointly, with the same total slot budget, backbone access, trainable parameter budget, and action-profile supervision. A *semantic embedding* control encodes the same candidate texts with the same reader. These distinguish latent content, candidate association, training, and ordinary semantic information.

Interlat and LatentMAS are evaluated with their original communication mechanisms, and with matched clarification controllers, clearly labeling the latter as adaptations [2, 13]. For representation-isolation comparisons, each communication mechanism exports the same slot budget to the common controller; decoder-native execution is reported separately because it can change proposal coverage. PropUQ-MAS supplies an inherited-risk representation, also with native and matched-control results; AUQ retains its explanation-guided native control [7, 12]. CAGE-Cal is compared only in its compatible final-answer panel setting, with all counterfactual graph calls charged [5]. It is not presented as a native controller for different subtasks.

All learned controls receive the same branch labels, training roots, compute-selection allowance, and final reporting calibration protocol. Shared menus are identical at identical public states, not forcibly identical after trajectories diverge. Generous-text and direct-rereading controls complement the matched-budget comparison. No claimed advantage can be attributed to exclusive tool access, a stronger responder, or a controller trained only for RCLUP.

5.3 RQ1: Does Propagation Limit Inherited Errors?

We replay the same naturally occurring prefix under each frozen receiver, initially without learned clarification. An erroneous-premise persistence (EP) audit tracks a reference-checkable entity–relation–value proposition from its sender to receiver and final answer. Native support/decomposition labels and deterministic date or calculation rules are used only after execution; unlabelable claims remain in overall task evaluation and are reported as unscorable for EP. Denominators are shown at every edge and terminal stage.

The mechanism comparison separates no dependency input, joint dependency features alone, features plus packet normalization, and full reception with clarification. Stratification uses graph depth, shared sources, and shared evidence-bearing ancestors. Duplicating an identical packet while fixing the textual state is a separate invariance check, not evidence of robustness to distinct correlated mistakes. Natural shared-source errors, not duplicate removal alone, must support a propagation claim. Where two legitimate downstream tasks consume the same upstream subproblem, matched handoffs test whether actual task conditioning changes the useful question; controlled examples are labeled separately when native pairs are unavailable.

5.4 RQ2: Are the Risk Targets Calibrated?

In fixed-prefix evaluation, every predictor scores the same commitment and realized π_0 continuation, holding answers and labels constant. We report Brier, Bernoulli NLL, error-positive AUROC, AURC, and reliability plots by role, action family, depth, and shared-source status. Calibration comparisons fix the learned logits and contrast no mapping, global logistic scaling, role-only scaling, and partially pooled role/action scaling. Ten equal-width bins define ECE; bin and group counts accompany it. A global score alone does not establish cross-role calibration.

A distinct on-policy evaluation uses the selected-action and terminal forecasts from Section 4.6, with actual π^* outcomes. No final-output score is attached to a pre-intervention label or used to change an answer. Macro-role Brier/NLL, worst-supported-group ECE, and unseen task-composition performance expose group failures concealed by pooled averages. Sparse groups use the prespecified global fallback, and their uncertainty is reported. Calibration is distribution-dependent; it does not establish message-level causal blame.

5.5 RQ3: Does the Latent Help Choose Clarifications?

An outcome-independent subset of saved evaluation prefixes is replayed with every admitted question and the common reference executor. All methods then face the same action outcomes. We measure selected-action utility, regret to the best measured menu action, premature stopping, and unnecessary questioning. Since an empirical maximum over noisy outcomes is optimistic, paired differences in regret and their intervals are primary. A second diagnostic compares the reference and executor-matched question heads to quantify the effect of retargeting.

The essential training ablation compares a commitment-only carrier frozen before question learning, joint commit/question BCE,

Table 2: Main closed-loop results at a two-item question budget. Each cell is unmeasured. Brier uses the actual final answer; Q is the mean number of atomic questions. Native and matched-controller (MC) rows are identified explicitly.

		HotpotQA			MuSiQue			DROP		
Model	Method	EM↑	Brier↓	Q↓	EM↑	Brier↓	Q↓	EM↑	Brier↓	Q↓
Qwen3-8B	Text MAS	—	—	—	—	—	—	—	—	—
	Action-trained scalar	—	—	—	—	—	—	—	—	—
	Rich text + MC	—	—	—	—	—	—	—	—	—
	Generic latent + MC	—	—	—	—	—	—	—	—	—
	PropUQ-MAS + MC	—	—	—	—	—	—	—	—	—
	Interlat + MC	—	—	—	—	—	—	—	—	—
	LatentMAS + MC	—	—	—	—	—	—	—	—	—
	AUQ, native	—	—	—	—	—	—	—	—	—
	RCLUP	—	—	—	—	—	—	—	—	—
Mistral-7B	Text MAS	—	—	—	—	—	—	—	—	—
	Action-trained scalar	—	—	—	—	—	—	—	—	—
	Rich text + MC	—	—	—	—	—	—	—	—	—
	Generic latent + MC	—	—	—	—	—	—	—	—	—
	PropUQ-MAS + MC	—	—	—	—	—	—	—	—	—
	Interlat + MC	—	—	—	—	—	—	—	—	—
	LatentMAS + MC	—	—	—	—	—	—	—	—	—
	AUQ, native	—	—	—	—	—	—	—	—	—
	RCLUP	—	—	—	—	—	—	—	—	—

Table 3: Separate propagation and calibration endpoints. EP is erroneous-premise persistence; NLL scores common fixed-prefix outcomes. Reporting-only recalibration cannot change EP. All cells are unmeasured.

Variant	EP↓	Brier↓	NLL↓
No dependency representation	—	—	—
Dependency features only	—	—	—
Full packet-normalized reader	—	—	—
Fixed reader, uncalibrated	—	—	—
Fixed reader, global calibration	—	—	—
Fixed reader, role-only calibration	—	—	—
Fixed reader, partially pooled	—	—	—

Table 4: Training and representation mechanisms. Question gain is measured under the common fixed executor; EM is closed-loop. All entries are unmeasured.

Variant	EM↑	Gain↑	Ask NLL↓
Full RCLUP	—	—	—
Commit-only carrier, frozen	—	—	—
Joint BCE, no profile differences	—	—	—
No executor retargeting	—	—	—
Rich text, actual task	—	—	—
Candidate latent, role name only	—	—	—
Rich text, role name only	—	—	—
Generic latent, joint supervision	—	—	—
Candidate semantic embeddings	—	—	—
No inherited-content forwarding	—	—	—
No learned clarification	—	—	—

and full BCE plus action-profile differences. All variants use the same carrier capacity and response labels. A factorial ablation crosses candidate-associated latents with actual receiver-task conditioning. Additional controls use role names alone, generic pooling, semantic embeddings, no recursive forwarding, no dependence features, and no clarification. Candidate–latent shuffling is an association diagnostic, not a main baseline. Receiver-specific question value is evaluated in continuation-risk strata, so improved action discrimination can be distinguished from a generally better failure score.

Question addressability, candidate/menu coverage, source validity, and unresolved replies are reported alongside utility. A missing useful question is a proposal failure that no selector can repair. Reading a known source, requesting an upstream interpretation, and issuing a new retrieval call are logged separately; a same-compute rereading control tests whether an apparent clarification benefit is simply additional processing.

5.6 Closed-Loop Outcomes, Cost, and Inference

Final EM, native token F1, available support scores, and actual-output calibration are evaluated at atomic-question budgets 0, 1, 2, 4. The main setting permits two items. Recovery and regression compare each method's final answer with the no-learned-clarification reference on the same question and seed; unresolved and failed executions count as errors. Each frozen budget-specific policy receives independent calibration. Tables 2–3 reserve the corresponding numerical results.

Costs include all sampled proposals, representation extraction, question generation, source responses, graph processing, checks, repair, and reporting heads. We record input/output tokens, latent slots and bytes, forward passes, latency, peak memory, and offline branch-label cost. Matched-question and matched-compute curves

are separate; neither a latent vector nor cached work is free. All methods share hardware and parallelism.

Selection uses development data only. Paired root-question bootstrap intervals use 10,000 resamples, with seeds and branches nested in each root. The primary family comprises RCLUP-versus-rich-text and RCLUP-versus-generic-latent EM contrasts on HotpotQA and MuSiQue for both backbones: eight contrasts receive Holm correction at 0.05. Secondary mechanism strata are descriptive when small. Non-significance is not evidence of equal quality. All failed runs and protocol amendments remain in the result manifest.

6 Scope and Responsible Use

The method requires hidden-state access and backbone-specific projections. Its candidate and question sets bound the distinctions it can use; an omitted interpretation cannot be recovered by selecting among the remaining items. Source lineage indicates dependence, not source truth, and normalization can discount genuinely useful overlapping evidence. Action risks are relative to their specified continuation; deployed-policy calibration remains distribution-dependent and is not an individual guarantee. Latent messages can retain sensitive source information and provide no privacy boundary. The evaluation uses benchmark evidence and simulated inter-agent exchanges, not human-benefit or live-Web safety claims. Source licenses, access controls, and human verification remain necessary.

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A Implementation Specification

A.1 Frozen Proposal and Evidence Interfaces

Each invocation collects eight stochastic proposals at temperature 0.7, top-p 0.9, plus one greedy proposal. Only stochastic draws enter the support counts. Keep four distinct candidates when available, with $\alpha = 0.1$ and a residual category. In Qwen3-8B the thinking mode is disabled. The per-call generation limit is 512 tokens; failures and truncations remain logged. The decoder and tool interfaces remain frozen throughout all learning stages.

Retriever artifacts are grouped by ordered source identifiers, interpreter artifacts by normalized entity–relation–value tuples and source identifiers, reasoner outputs by native public answer normalization, and verifier outputs by verdict plus normalized correction. Ordered fields remain ordered when meaningful. Invalid outputs enter residual mass rather than being silently retried; normal repair determines any retry. Exact canonicalization can over-separate paraphrases, so fragmentation and candidate coverage are measured. The greedy candidate may have zero stochastic count. All methods use the same grouping and retained set at a shared public state.

Sources are keyed by document version and quoted offsets. An evidence-bearing ancestor is an invocation whose artifact or cited source is consumed; the planner, scheduling nodes, and the original user question are not counted as common evidence ancestors. Packet identity is (origin, version, content hash). Conflicting content under one origin/version is an integrity failure, not an opportunity to average claims. Source or ancestor overlap is computed only from known identifiers; empty–empty Jaccard overlap is zero. Unknown provenance has an explicit mask. Readable-history encoding uses the canonical packet list; exact-packet multiplicity is not reintroduced through duplicated text.

A source-backed reply contains `answer`, `source_ids`, `spans`, `unresolved`, and `evidence_event_id`. Quote offsets must match the cited snapshot. A failed quotation check marks that evidence unsupported and passes the issue to the common verifier. Neither a matched quote nor a model's confident answer establishes semantic correctness. New valid or unresolved observations update history; only supported observations supply support. Previously consumed independent sources remain available on revision. Repeated unresolved questions against the same source version are masked. Changed source versions reopen the question. No hidden answer is supplied to the sender.

A.2 Module Dimensions and Dependency Features

The final backbone layer supplies response states. The pooler has $m = 4$ learned queries per candidate, width $d = 256$, and independent key/value projections. Four tied residual slots use a missing-content flag. Text, task, question, and action embeddings are mean-pooled final-layer states projected to d. Two four-head, pre-layer-normalized attention blocks have width-4d GELU feed-forward layers. Commitment, question, packet-gate, and reporting heads each use hidden widths 256 and 128 with GELU and dropout 0.1. Modules are shared across roles within a backbone and fitted separately across backbones. The candidate feature includes its actual text, not just an ordinal slot.

For slots with origin o, evidence-ancestor set A, and source-span set P,

$$d_{uv} = [\mathbb{I}[o_u = o_v], \mathbb{I}[o_u \in A_v], \mathbb{I}[o_v \in A_u], \\ \text{Jaccard}(A_u, A_v), \text{Jaccard}(P_u, P_v), m_{uv}]. \tag{11}$$

The packet summary d_i contains its mean and maximum overlaps, number of known origins, and missingness. The public graph vector v_j contains edge count, unique-origin count, depth, maximum overlap, role counts, candidate-support entropy, and remaining budgets. Root reads use $c_{jb} = 0$ with a no-parent mask. The candidate-reading weights in Equation (5) are attention weights, not correctness probabilities. All encoder, pooler, gate, and commitment parameters belong to θ; question heads use κ_0 or κ, and the reporting head uses ψ.

B Action Profiles and a Fixed Training Schedule

B.1 Data Partitions and Collection Bounds

Official training roots are assigned by a fixed SHA-256 hash of dataset and root/composition identifier to five non-overlapping partitions: representation fitting 55%, policy-risk fitting 15%, selection 10%, action calibration 10%, and policy calibration 10%. The hash salt is `rclup-v2-protocol`. From these partitions, hashes select respectively 256, 128, 128, 128, and 128 roots per dataset. If a partition is smaller, all its roots are retained and the realized count is reported. Official labeled development questions provide 512 held-out evaluation roots selected by the same outcome-independent ordering, or the whole split when smaller. No test metric selects a sample, checkpoint, threshold, or source exclusion.

At most two receiver prefixes per fitting root are selected by role-balanced hashing, with all their ancestors retained for carrier reconstruction. Half of the prefix pool is generated by the ordinary π_0 workflow and half by a fixed round-robin menu policy admitting up to two questions, chosen without outcome information. Post-reply states are also retained for commitment training. Missing prefixes remain missing rather than being replaced by favorable outcomes. All candidates, replies, and seeds from one root stay in that root's split. Three module-fitting seeds are 13, 29, and 47; branch seeds are 101, 211, and 307.

B.2 Phase A: Learn the Carrier from Both Actions

The immutable reference policy π_0 comprises the public planner and role prompts, stochastic backbone commitments at temperature 0.7 and top-p 0.9, tools, and two ordinary verifier-repair rounds. It receives readable candidates and sources but no trained latent adapter. For commitment labels, override only the current artifact, then run π_0. For reference-question labels, execute the admitted question, obtain an actual source-backed reply, rebuild affected public states, and let π_0 choose the post-reply artifact and remaining suffix. Every admitted action uses all three predetermined branch seeds. Invalid or unanswered replies are not discarded.

The collection module stores only frozen-backbone traces and public state before an action in that action's predictor input. Its

eventual reply and suffix can label the action, but cannot enter those features. During each training pass, candidate latents are recomputed in ancestry order, with gradients through the complete retained ancestry. The current parameters never affect the fixed executors used to generate Phase-A labels. Each root receives total loss weight one; its prefixes, commitment candidates, questions, and seeds are averaged within the separate terms of Equation (7). Profile differences include all unordered admitted-action pairs; a set with fewer than two actions contributes zero profile loss. The default is $\mu = 0.5$.

B.3 Phase B: Retarget to the Frozen Learned Receiver

Freeze the selected θ^*, including the commitment head and all carrier construction. For every saved fitting question and seed, produce or reuse a byte-identical sender response, reconstruct its revised state, select Equation (8), and execute π_0 thereafter. Fit g_κ from g_{κ_0} on these executor-specific final-error outcomes by root-weighted BCE. The question features are the original pre-reply features. Gradients do not update the carrier in this phase: its direct clarification training occurred in Phase A. An ablation without retargeting deploys g_{κ_0} unchanged, but is not described as estimating executor-matched risk.

Candidate commitments retain their Phase-A targets because the committed artifact and remaining π_0 are unchanged. Cached suffixes are reused only if the complete prompt, model, seed, tool definitions, source snapshot, and committed artifact match. A reply reused across heads is not a free response at deployment. Reference-question predictions are used only in fixed-executor comparisons; deployed questions use g_κ.

B.4 Phase C: Action Calibration and Policy Freezing

Both commitment and executor-matched question outcomes are collected on action-calibration roots. The default group-deviation penalty in Equation (9) is 1, and at least 50 distinct prefixes are required for a group-specific deviation. Smaller groups use the global map. The optimizer minimizes root-weighted Bernoulli NLL plus the penalty. The role-only and global controls use identical logits and labels. Fixed-executor question diagnostics fit a separate calibration map on reference-question outcomes from these same roots; that map is never substituted for executor-matched calibration. Group-specific positive slopes preserve the within-role commitment ranking, so calibration cannot change Phase-B's reply executor.

The penalty grid is $\lambda \in \{0, 0.01, 0.02, 0.05, 0.10\}$. Under each fixed hard question/inference budget, the selected value maximizes selection EM after calibration on its own partition. Ties prefer fewer questions and then smaller λ. Representation and head checkpoints have already been selected by their respective NLL criteria; λ selection does not change them. Calibrators never use selection outcomes as calibration labels. The learned receiver, question head, action calibrator, proposer, sources, and budget configuration are then frozen as π^*.

B.5 Phase D: Forecast the Actual Executed Trajectory

Run π^* on policy-risk-fitting roots, which trained neither the carrier nor the question head. Each selected pre-action record contains its role, action, state, budget, and frozen latent features; the final record contains the actual answer and final state. No pre-action record contains a future reply or final answer. All records are labeled with the final error of that same rollout. Fit the reporting head h_ψ by root-weighted BCE. Use the independent policy-calibration roots to fit a partially pooled logistic map with groups (role, phase, action family), phase distinguishing selected-action and final-output records. The final group has its own phase identity. Its default penalty and support threshold equal Phase C's. Reporting is never an input to the controller. Repeat this phase for each frozen question-budget policy.

B.6 Optimization and Accounting

Use AdamW, learning rate 10^{-4}, weight decay 10^{-2}, batch size 64 prefixes, gradient clipping at 1, and at most ten epochs. The representation checkpoint minimizes selection action-profile NLL (equal weight for commit and ask); deployment and reporting heads minimize their own selection NLL on the corresponding executor. Policy-risk-fitting roots are not reused to choose the controller. The fixed search is a learning-rate grid $\{3\times10^{-5}, 10^{-4}, 3\times10^{-4}\}$ with all architecture dimensions unchanged; every learned control gets the same three runs and seeds. The profile coefficient $\mu = 0.5$ is fixed, with $\mu = 0$ an ablation, not test-based selection.

Collection is expensive despite the small trainable modules. With at most five commitments, six questions, and three seeds, one selected prefix requires at most $3(5 + 6 + 6) = 51$ reference/retargeted suffix executions before calibration and policy reporting. The 256-root, two-prefix cap permits up to 26,112 such suffixes for one fitted receiver. Sharing Phase-A labels across three fitting seeds gives $16{,}896 + 3(9{,}216) = 44{,}544$ base suffixes per dataset and backbone, before additional post-reply records. This is a collection bound, not an observed cost or an instruction to start paid runs. An eight-prefix implementation check measures actual work before collection; any reduced budget is frozen in a dated amendment before test evaluation, with the same adjustment applied to all methods.

C Workflow, Comparisons, and Evaluation Contracts

C.1 Shared Tools and Finite Execution

The source pool is the complete released per-question candidate-document set, never a pool selected by gold support. BM25 uses $k_1 = 1.2$, $b = 0.75$, returns three passages per query, and permits direct fetch by ID. A workflow has six normal retrieval queries and up to two verifier repairs; clarification-triggered retrieval consumes the same query budget. The default learned-clarification budget is two atomic items, swept over zero, one, two, and four. Menus contain at most six questions, deterministically ordered by sender and candidate-pair support. Missing alternatives are addressed by candidate-versus-residual scope questions. Schema filtering happens before outcome generation; invalid menus are retained as coverage failures.

The public planner forms at most three retriever/interpreter branches and a final reasoner/verifier sequence. Every regeneration consumes a global limit of 64 role invocations. Shared maximum output/input processing budgets are 65,536/524,288 tokens per workflow. Ordinary source access is not forbidden to make communication helpful. Budget exhaustion yields the best checked artifact or unresolved; it never removes the run or truncates necessary history. New messages invalidate descendant states, including compressed representations and risk scores, by origin/version. Re-execution uses the same rules for all methods. This information-task protocol does not claim to reverse external real-world actions.

C.2 Matched Representations and Task-Conditioning Tests

The rich-text sender supplies up to 128 explanation tokens per retained candidate, including the candidate's uncertainty source and supporting/contradicting spans. A generous-text control permits 512. The same frozen backbone produces these explanations without a gold answer; their cost is charged. Generic latent uses the same $(K_i + 1)m$ slot capacity and all candidate traces without candidate-to-slot association; readable candidate identities remain available. Total learned parameter counts are matched within 5% by a width adjustment recorded before evaluation. A semantic-embedding arm replaces hidden response traces with encoded candidate/explanation text. All representations receive the same provenance and both joint losses.

Task-conditioning ablations change only inputs to the trained receiving/control module. Ordinary agents retain their assigned task, tools, and complete allowed context; the role-name-only control is not a crippled agent. Natural task-sensitivity pairs require a common upstream subproblem legally consumed by distinct downstream operations. Their count and provenance are reported. Test-time latent shuffling preserves candidate texts, weights, and slots, and is reported as an association diagnostic only.

C.3 Risk and Control Endpoints

For error probabilities $\hat{r}_n$ and errors e_n, Brier is the mean $(\hat{r}_n - e_n)^2$ and NLL is mean Bernoulli negative log likelihood. Clip only for numerical NLL at 10^{-6}. AUROC takes errors as positive. AURC integrates error among examples retained in ascending risk order. Ten-bin ECE is complemented by bin counts and calibration curves. High-confidence error (HCE) counts errors assigned correctness at least 0.9 over all eligible records; report also its error-conditional denominator. Roots, not sampled suffixes, are the independent units.

For fixed-executor menu diagnostics,

$$\bar{U}_j(u) = 1 - \frac{1}{R}\sum_r e^0_{jur} - \lambda c(u), \qquad c(b) = 0, \tag{12}$$

with regret $\max_u \bar{U}_j(u) - \bar{U}_j(\hat{u}_j)$. The same action realizations are used for all selectors. The empirical maximum is an optimistic finite-sample reference, not the Bayes-optimal information policy. Gain signs near zero are reported with uncertainty rather than treated as exact ground truth. Candidate/menu coverage, addressability, quote validity, and source-backed answerability are distinct measurements. Selected policies with different reply executors have their own labels and are evaluated separately end to end.

Natural EP requires the same normalized proposition, including entity and scope, to be reference-checkable upstream and downstream. When native data do not provide that label, the instance is unscorable for EP, not pseudo-labeled by an LLM. Its final task result remains. Recovery and regression use all originally wrong and originally correct reference outcomes respectively as denominators. EM/F1 and native support evaluation remain separate where the dataset does not define a validated combined grounded-success label. DROP's paragraph supports the arithmetic experiment, not open-Web retrieval.

C.4 Boundary Checks

Three mathematical checks clarify the implementation. First, for a fixed role and commit family, the map $\sigma(e^{a_0+a_g}\ell + b_0 + b_g)$ is strictly increasing in ℓ; therefore its argmin equals the raw-logit argmin. Action calibration cannot silently change the post-reply executor. Second, Phase-A outcomes depend on the frozen π_0 and actual observed replies, not on θ, so question gradients do not create moving labels. Third, exact packet duplication is invariant after canonical coalescing when the canonical readable state is fixed. None of these facts establishes robustness to distinct correlated misinformation; that claim requires RQ1's natural-error results.

The reporting head has no control path. Recalibrating it must leave selected artifacts, questions, and final answers unchanged. Reference and executor-matched action maps, on-policy reporting maps, and source snapshots have separate version keys. All metrics use the score corresponding to the actual stated event and policy. These checks are specifications for the implementation, not claims that a full MAS has already run.

D Additional Results

Table 5: Role-wise executed-policy calibration. Columns show Brier before and after reporting-only calibration and NLL after it; N is the number of held-out roots contributing records. All entries are unmeasured.

Role/phase	N	Brier raw	Brier cal	NLL cal
Retriever	—	—	—	—
Interpreter	—	—	—	—
Reasoner	—	—	—	—
Verifier	—	—	—	—
Final answer	—	—	—	—

Table 6: Budget and cost results for the fixed RCLUP policy family. Latency, tokens, and latent bytes include all communication and repair overhead. Numerical results are unmeasured.

Budget	EM	Items	Tokens	Latent bytes	Latency
0	—	—	—	—	—
1	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—